Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/30/2021
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 9/16/21) NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 8/4/2020) NOTE: End date changed to 10/31/2020 per F. Hernandez/ARC (Ed., 6/20/19) NOTE: End date changed to 10/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 10/31/2018 per F. Hernandez/ARC (Ed., 9/21/17) NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	NOTE: Per the PI's technical report, Dr. Sharmila Bhattacharya is no longer with the project as Co-Investigator (Ed., 7/7/23). September 2020 report: (1) Jennifer Chou (technical assistant) rejoined the project in July 2020. August 2021 update: (2) PhD student Brian Wey defended his thesis on Oct 14, 2020 and graduated Feb 2021.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB42G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP (virus-like particles) genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Parasitic wasps of Drosophila are a powerful and increasingly well-characterized model for understanding longstanding questions about natural host/parasite interactions, allowing molecular genetic examination of both innate immune mechanisms in fly hosts and virulence mechanisms of their wasp parasites. Distant relatives of these parasitic wasps are used for biocontrol of agricultural pests worldwide. August 2021: The "Earth benefits" impact of our research over the last year has been as follows: (a) RNA-seq analysis of adult fruit flies (wild type and mutants developing inflammatory microtumors) will help in understanding the effects of spaceflight in humans and develop countermeasures for immune dysfunction. (b) RNA-seq analysis of parasite transcriptomes in response to spaceflight with emphasis on virulence-related genes is providing novel insights into how endoparasites (those that develop within their hosts) react to the pressures as they develop in low Earth orbit. (c) Eight students were trained over last year. One technician also contributed to the project. Two Federal Work Study students assisted and contributed to the project. (d) Two peer-reviewed papers were published in the last year. One oral presentation was made in the Annual Drosophila Research Conference in spring 2020. These presentations helped disseminate our research results.
Task Progress:	Space exploration presents significant challenges to human physiology, affecting all biological systems. Effects of spaceflight on human immunity are of particular interest, especially in the context of exploration-class missions. A number of studies have focused on immune dysfunction in space, but significant gaps remain in our understanding of the specific effects of spaceflight and microgravity on the innate immune system at the cellular and molecular-genetic levels. These include how different signaling pathways affect immune physiology and their links to development and survival in space. Drosophila melanogaster is a simple and genetically accessible model system, deficient of most hallmarks of adaptive immunity. The seminal discoveries of the Toll-like receptors and Toll-NF-kappaB signaling as central mediators of innate immune responses in mammals were catalyzed by earlier genetic studies in Drosophila. More recently, studies have uncovered the effects of spaceflight on immune gene expression, blood cell dysfunction, and host competence against microbial infections. To better understand the effects of spaceflight on host-parasite interaction, natural parasitic wasps of Drosophila were sent to the International Space Station (ISS) in April 2018. The parasites attack the larval stages of fruit flies. After laying eggs in fly larvae, they hatch and eat the developing fly host, and emerge from the pupal case in about 20-25 days. The overarching goals of the study were: First to examine if the success of the fly/wasp partner is altered in space (who wins the "arms race"); and second, to determine, if parasites reared in space have the same virulence (or reduced virulence) as their counterparts on Earth. We and others have recently shown that parasitic wasps of Drosophila produce particles in their venom that specifically block the cellular immune arms of their hosts. These particles are most similar to eukaryotic extracellular vesicles, although they possess an unusual spiked morphology, more similar to viruses. We refer to them as multi-strategy extracellular vesicles (MSEVs). We have identified gene expression changes in both the hosts and the parasites (including genes encoding MSEV proteins) due to spaceflight conditions and are in the process of understanding the trends in these changes.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Govind S, Chou J, Heavner M, Ramroop J, Razzak Z, Wey B. "Drosophila and its parasitic wasps: Understanding the host-parasite interface." 62nd Annual Drosophila Research Conference, Virtual, March 23–April 1, 2021. 62nd Annual Drosophila Research Conference, Virtual, March 23–April 1, 2021. A Dros Res Conf. 129 abstract. , Mar-2021
Articles in Peer-reviewed Journals	Ramroop, J, Heavner, ME, Razzak, ZH, Govind, S. "A parasitoid wasp of Drosophila employs preemptive and reactive strategies to deplete its host's blood cells." PLoS Pathog 2021 May;17(5):e1009615. https://doi.org/10.1371/journal.ppat.1009615 ; PMID: 34048506; PMCID: PMC8191917 , May-2021

Articles in Peer-reviewed Journals	Paul AM, Mhatre SD, Cekanaviciute E, Schreurs A-S, Tahimic CGT, Globus RK, Anand S, Crucian BE, Bhattacharya S. "Neutrophil-to-lymphocyte ratio: A biomarker to monitor the immune status of astronauts." Front Immunol. 2020 Nov 2;11:564950. https://doi.org/10.3389/fimmu.2020.564950 ; PMID: 33224136; PMCID: PMC7667275 , Nov-2020
Dissertations and Theses	Wey B. "Insights into Leptopilina Spp. Immune-Suppressive Strategies Using Mixed-omics and Molecular Approaches." PhD thesis, Graduate Center, City University of New York, February 2021. https://academicworks.cuny.edu/gc_etds/4162/ , Feb-2021
Dissertations and Theses	Sevilla CM. "Molecular and genetic studies on a SipD/IpaD-related virulence spike protein from immune-suppressive particles of a Drosophila parasitoid." MS Thesis, The City College of New York, CUNY, June 2021. https://academicworks.cuny.edu/cc_etds_theses/916/ , Jun-2021
Significant Media Coverage	Govind S. "PI's presentation (Drosophila and its parasitic wasps: Understanding the host-parasite interface) at 62nd Drosophila Research Conference included with Speaker Highlights." Plenary Session Presentation Speaker Highlights from Annual Drosophila Research Conference. https://genestogenomes.org/dros21-speaker-highlights/ , Mar-2021